

MatMul-Free Model: Performance Analysis

Training Snapshot

Checkpoint	Step 9,000 / 10,000
Model Size	~257M parameters
Dataset	SlimPajama-6B
Hardware	4x NVIDIA A10G

Key Performance Metrics

-53.5%

Throughput vs. Baseline

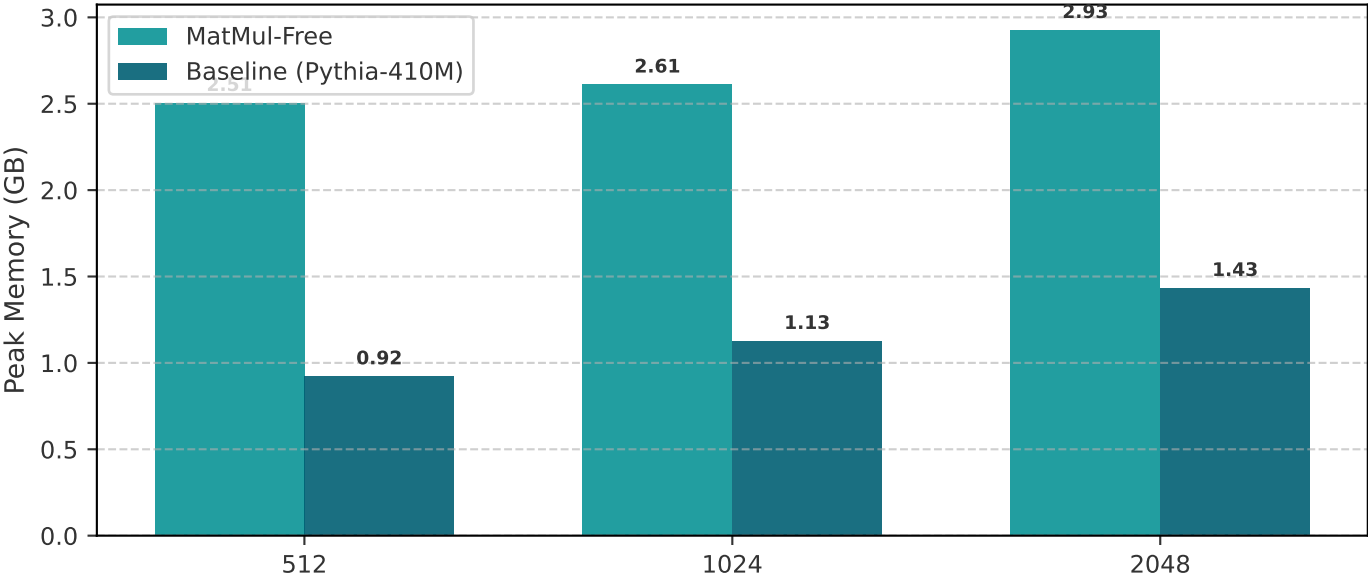
Higher is better

+104.1%

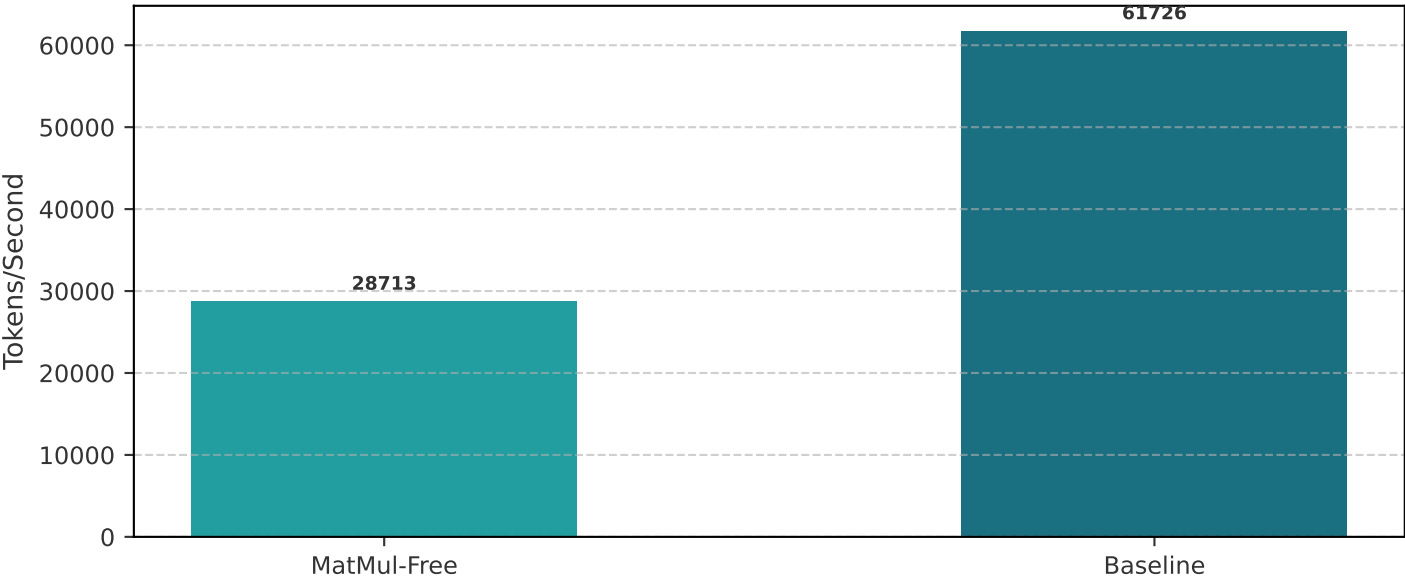
Memory Overhead (seq=2048)

Lower is better

Memory Usage Comparison

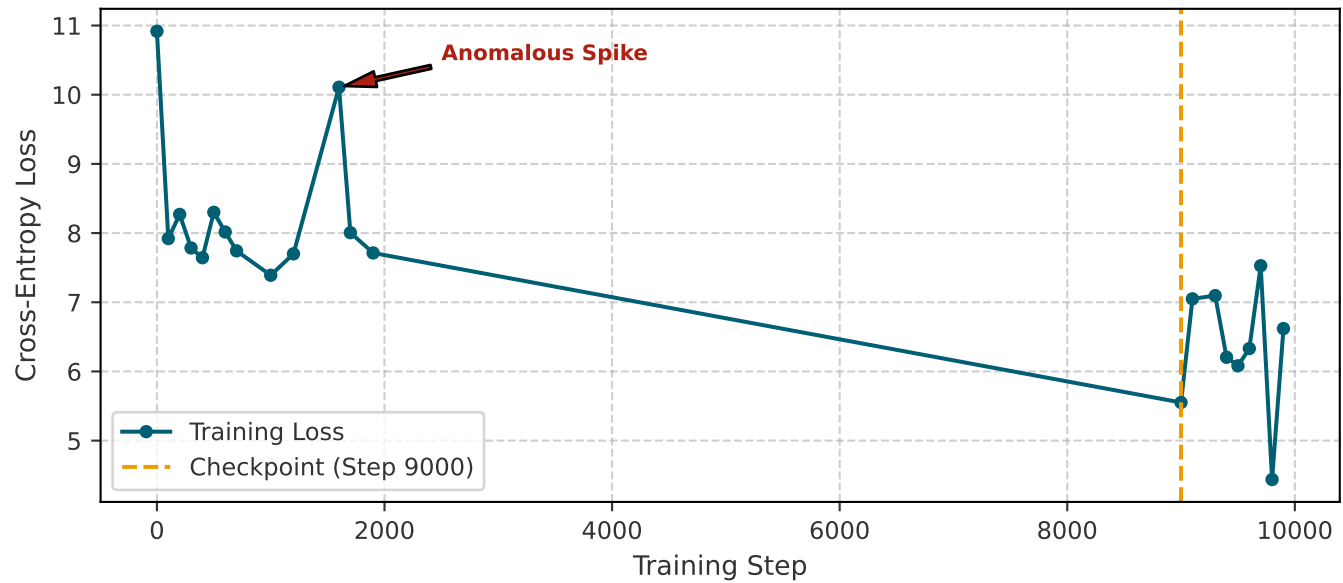


Throughput Comparison (seq=1024, batch=4)

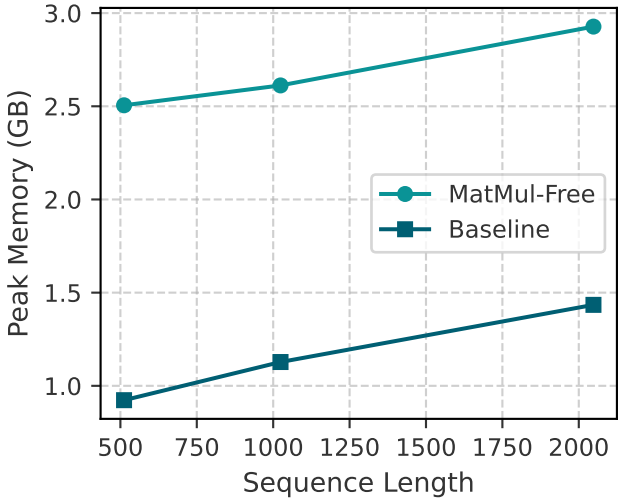


Training Trajectory & Memory Analysis

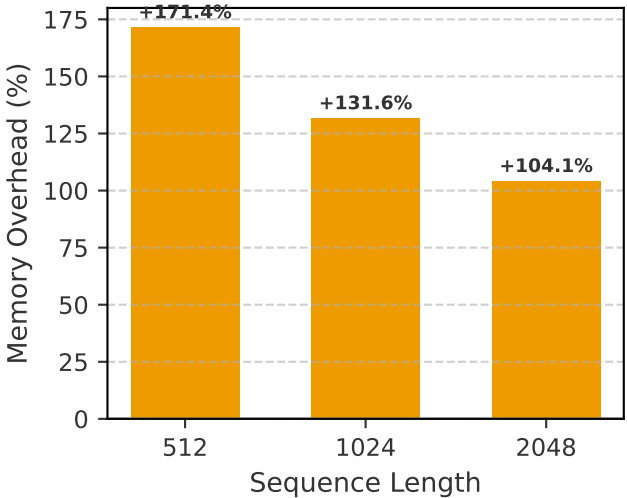
Training Loss Trajectory



Memory Scaling



Overhead vs. Baseline



Implementation Notes

- **Quantization:** Weights are 1.58-bit (Ternary: -1, 0, 1), Activations are 8-bit. - **Attention:** Standard Softmax attention is replaced with a Hierarchical Gated Recurrent Network (HGRN). - **Fused Kernels:** Triton is used to fuse LayerNorm, Linear, and Quantization operations, reducing overhead. - **Triton Autotuning:** Disabled in this fork due to compatibility issues, using fixed default kernel configurations.